

PARTS LIST

Semiconductors:

LED1, LED3	- IR transmitting LED
LED2, LED4	- Photodiode/IR receiving LED
IC1	- LM358 operational amplifier
D1, D2	- 1N4007 rectifier diode
LED5, LED6	- 5mm LED
T1-T2	- BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R8	- 100-ohm
VR1, VR2	- 10-kilo-ohm preset

Miscellaneous:

BATT.1, BATT.2	- 9V battery
RL1, RL2	- 9V, 1CO relay
M1, M2	- 10rpm, 9V geared motor
	- Robot cabinet/chassis
	- Two robot wheels
	- Small pedestal wheel
	- Jumper wires
	- 2-pin connector for battery



Fig. 4: Module with IR sensor pair and LM358

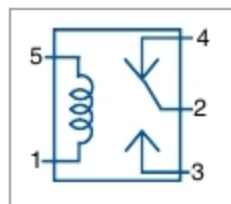


Fig. 5: Pin diagram of the relay

connected in reverse-biased mode across the relay for protection.

White surface.

For white surface, output of the comparator connected to IR sensor is high, because IR ray reflects on a white surface. Output of the comparator drives the transistor and, hence, transistor conducts and current flows through the relay coil and the relay is energised.

Black surface. Output of the comparator connected to IR sensor is low (0V) for a black surface. The transistor does not conduct and no current flows through the relay coil. Thus, relay is in de-energised state.

Black line follower robot

The circuit diagram of the black line follower robot is shown in Fig. 3. When the right side of the robot is on the white surface, the right motor is activated. When the left side of robot is on the black surface, the left motor is inactive. Hence, the robot turns left.

When the right side of the robot is on the black surface, the right motor is inactive. When the left side of the robot is on the white surface, the left

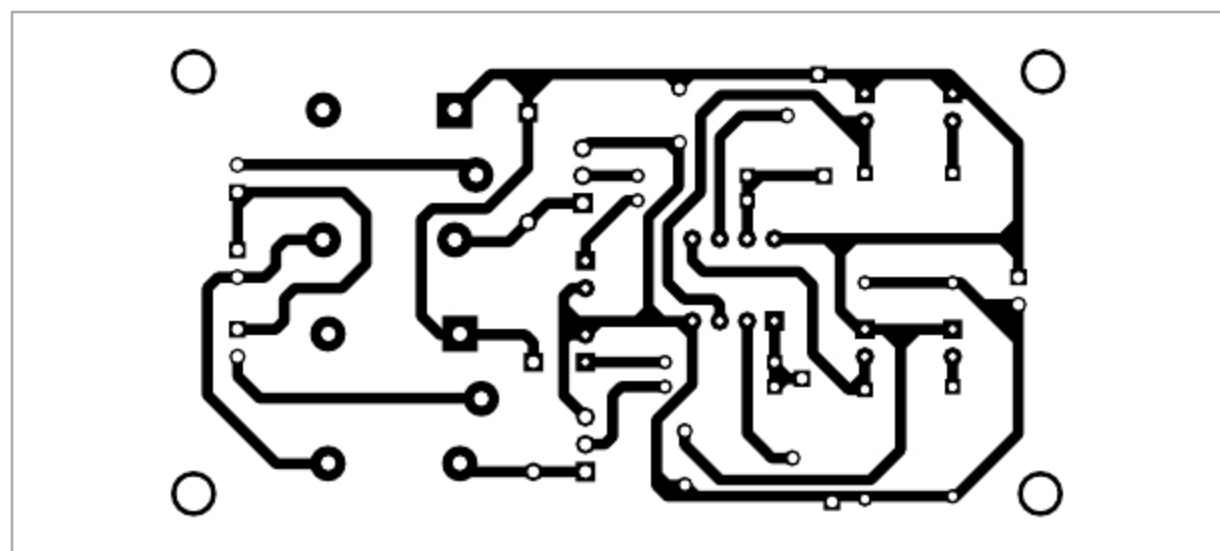


Fig. 6: Actual-size PCB layout of line follower robot

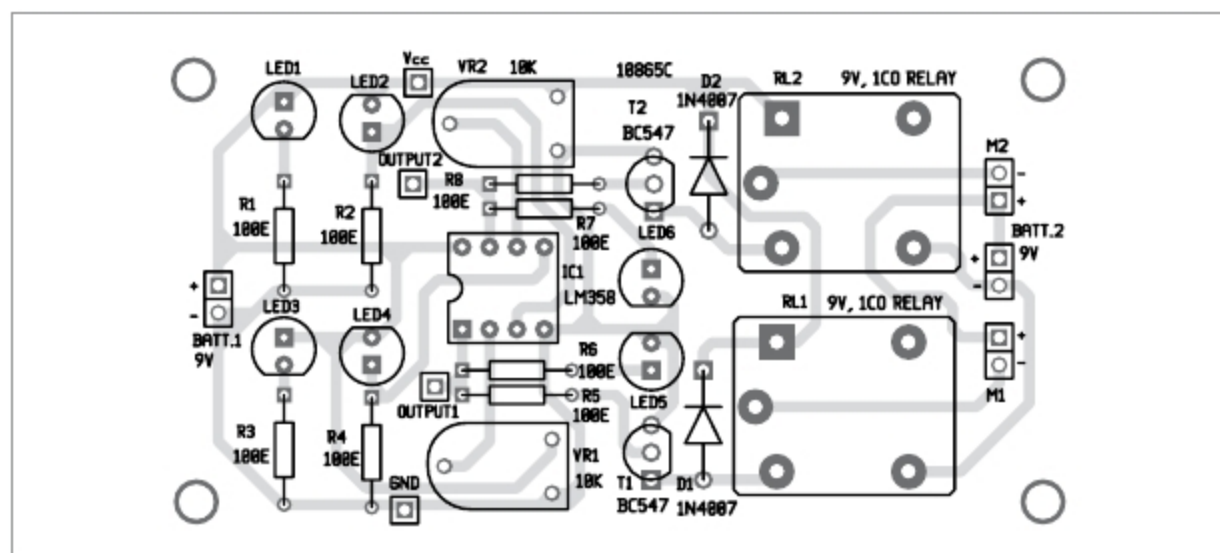


Fig. 7: Components layout for the PCB

motor is activated. So, the robot turns right.

When both sides are on the white surface, both motors are activated. And, the robot goes straight.

White line follower robot

The circuit diagram of the white line follower robot is the same, except the changing NO and NC contacts of the relay. When the relay is de-energised, pins 4 and 2 of the relay (Fig. 5) are shorted. Hence, current flows through the motor.

When the right side of robot is on the white surface, the right motor is inactive. When the left side of the robot is on the black surface, the left motor is activated. So, the robot turns right.

When the right side of robot is on the black surface, the right motor is activated. When the left side of the robot is on the white surface, the left motor is inactive. So, the robot turns left.

When both sides are on the black surface, both motors are activated.

Hence, the robot follows a straight line.

Construction and testing

An actual-size PCB layout of the relay-based line follower is shown in Fig. 6 and its components layout in Fig. 7. After assembling the circuit on the PCB, connect 9V battery to BATT.1 and BATT.2. Use jumper wires to connect IR TXs and photodiodes IR RXs. Place these properly on the chassis of the robot.

When you keep the robot on the floor, reflected rays from IR TXs should fall on photodiode IR RXs. Align the sensor pairs (IR TX and IR RX) in such a way that the robot follows the prefix black or white line. **EFY**



Souvik Kumar Das is passionate about electronics and MCU-based system design



Shibendu Mahata is M.Tech (gold medalist) in instrumentation and electronics engineering from Jadavpur University. His interests include designing MCU-based real-time embedded signal processing and process control systems